

# REQUIREMENT ANALYSIS

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## 1. Introduction

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### 1.1. Purpose of the system

This document defines the specifications and expectations of the development of a videogame, tentatively called "IFP-Tanks". The game is created for the advanced practical "Agile development of computer games" of Paul Brinkmann, Gero Brunke and Nils Schlaback in University Heidelberg.

### 1.2. Scope of the system

The scope of the video game encompasses the design and development of an top-down game in which the player is controlling a tank needing to eliminate all enemy tanks. After completing this objective it repeats in a new level with new enemys and new map design. The game is programmed using the game engine Godot with the language C#.

Key elements of the game include:

- Multiple individually designed levels grouped in worlds
- Multiple gimmicks for each world like switches and gates, teleporter and destroyable walls
- A variety of enemy tanks with different abilities
- Upgrades for player controlled tank
- Pixelart graphics

### 1.3. Objectives and success criteria of the project

The objectives of the project are:

- Try out game-specific software development processes in a "real laboratory" with a team of ideally 3 members.
- Learn how to use and extend a commercial game engine (Unity, Unreal Engine, Godot or others).
- Learning how to define an appropriate architecture for a given game that has the characteristics like flexibility, modularity etc.
- Efficiently solve a large software project in a team.

The success is measured by following criteria:

- The individual steps are well documented such as use-case/game storyboard, requirements analysis, architecture, documentation (including that of the tests), correct use of tools such as Git (version control, automatic testing with unit tests), Trello and correct administration with Kanban board.
- The extent the goal of agile development of a game has been implemented.
- The quality of the report(): clear, understandable structure, legible, correct, appropriateness of language?

## 1.4. Definitions, acronyms, and abbreviations

| Definition | Explanation                                                                                     |
|------------|-------------------------------------------------------------------------------------------------|
| Godot      | Game engine used to create video games                                                          |
| NPC        | Non-Player-Character: Entity controlled by the system                                           |
| HP         | Hit-Points                                                                                      |
| SFX        | Sound effects                                                                                   |
| HUD        | Heads Up Display: UI overlay, which is displayed in-game                                        |
| FPS        | Frames per Second: Unit of Framerate                                                            |
| GFX        | Graphical effects                                                                               |
| Pixel Art  | A style of digital art using pixelated graphics, often associated with retro gaming aesthetics. |

## 1.5. References

- GitHub: <https://github.com/nilssck/IFP-Tanks/>
- Miro: [https://miro.com/app/board/uXjVNcPgUHU=?share\\_link\\_id=453941137888](https://miro.com/app/board/uXjVNcPgUHU=?share_link_id=453941137888)
- Trello: <https://trello.com/b/wXEjhkif/orga>
- GitMind: <https://gitmind.com/app/planets/n3a0lcg>

## 1.6. Overview

The project "IFP Tanks" is created for the advanced practical "Agile development of computer games" and aims to deliver an top-down game where the player controls a tank and needs to eliminate all enemy tanks. This document provides initial outlines for scope, success criteria and requirements.

# 2. Proposed system

## 2.1. Overview

The game aims to deliver an immersive gaming experience where players control a tank, navigating through various levels to eliminate enemy tanks. The game emphasizes strategic combat and maneuvering skills as players progress through various levels, each presenting unique challenges with diverse enemy behaviors and map designs. With an intuitive control scheme and engaging pixel art visuals, the game offers an adrenaline-filled adventure that combines action, tactics, and progression.

## 2.2. Functional requirements

### 2.2.1. Player

The player controls a tank in a top-down view which can be [upgraded](#) and equip different [guns](#), to shoot with.

### 2.2.1.1. Movement and controls

The player has the ability to move his tank in all four directions using the **wasd**-Keys (**w**: move up, **s**: move down, **a**: move left, **d**: move right). He can aim the weapon of his tank using the **mouse**, which position is displayed by a recticle, and shoot the weapon with left-click at the aimed position. The camera is stationary and shows the entire playing field.

### 2.2.1.2. Stats

The tank controlled by the player has different stats which affect gameplay:

- **HP**: Hitpoints describe the current health of a player. He starts with 100HP. Once they reach 0 the player dies and is reset back to the last **checkpoints**.
- **MaxHP**: The players current HP can not exceed the maximum HP.
- **Damage**: The damage that the players weapon deals to enemies HP.
- **Rate of Fire**: The rate at which the player can shoot his weapon. It is measured in shots per second (sps).
- **Speed**: The players movement speed to navigate through the map.

### 2.2.1.3. Upgrades

The player can use coins, which are dropped by enemies when they get destroyed to buy upgrades in the **shop** after each level. Every upgrade becomes more expensive with every purchase, except Healing. Following upgrades are available:

- **Healing**: Allows to regenerate a certain amount of HP. HP can not exceed MaxHP.
- **Upgrade MaxHP**: Increases the maximum HP by x.
- **Increase Damage**: Increases the damage dealt by the player by x.
- **Speed**: Increases the movement speed by x.

### 2.2.1.4. Weapons

There are a number of different weapons which are acquirable through the **shop** using coins aswell. These weapons differ in their shooting mechanics and damage statistics. Following weapons are available:

- **50cal**: This gun shoots a straight flying shell which gets destroyed by hitting a wall.
- **Rocket launcher**: The rocket launcher shoots a rocket which homes to an nearby enemy tanks position by updating its target location several times. The rocket itself cannot dodge and is destroyed as soon as it hits a wall.
- **Laser**: The laser fires, after a short charging time, in a straight line damaging all enemies in it's path but it stops when hitting a wall. The "projectile" of the laser has no travelling time.
- **Grenade launcher**: The grenade launcher fires in an arc allowing the grenades to travel over a wall and damage all enemies in a small damage radius.

### 2.2.1.5. Damage and death

The player takes damage when hit by an enemy projectile. It is calculated by following formula:  $\text{NewHP} = \text{OldHP} - \text{Damage}$ . Once the HP of the player reaches 0 he will die and the game is reset to the last checkpoint.

## 2.2.2. Enemies

There are different enemies, continuatively referred to as "tanks", each having different stats, abilities and behaviours. They differ by their color so they can be easily differentiated.

### 2.2.2.1. Types

- **50cal (Blue)**: This tank shoots a straight flying shell at the players position, when in sight. The shell gets destroyed by hitting a wall.
- **Bouncing (Red)**: This tank shoots a straight flying shell at the players position, which bounces away when hitting a wall and gets destroyed when hitting a wall the second time.
- **Laser (Yellow)**: This tank shoots a laser after a brief charging period which behaves the same like the players laser.
- **Kamikaze (Black)**: This tank drives straight to the player exploding on impact or when it gets destroyed. It has no weapon equipped.
- **Rocket (Green)**: This tank uses a rocket launcher, that behaves like the players rocket launcher.
- **Invisible (White)**: This tank is invisible by default and only appears for a brief period after shooting a shell like the 50cal at the player.

### 2.2.2.2. Stats

Like the player, each tank has it's own of the following stats:

- **HP**: Hitpoints describe the health of a tank. Once they reach 0 the tank dies and is removed from the map.
- **Damage**: The damage that the tanks weapon deals to the player.
- **Rate of Fire**: The rate at which the tank can shoot its weapon. It is measured in shots per second (sps).
- **Speed**: The tanks movement speed to navigate through the map.

## 2.2.3 Levels and progression

The game is structured into levels which are grouped into worlds. Each world introduces new enemies and tiles mixing up the gameplay.

### 2.2.3.1 Level

A level is a distinct, individual section of the game. Following points characterize a level:

- **Environment**: An arrangement of specific tiles on the grid. Each level is created and designed by hand.
- **Enemies**: Enemies can be placed freely on the grid. They can not be placed on wall-tiles.
- **Objectives**: The objective of the player is to kill all enemies in the level. Once all enemies are defeated, the level is completed and the shop is opened.

#### 2.2.3.1.1 Grid design

Each level is divided into a grid. On each cell a tile must be placed. Each level is hand-crafted by placing all tiles manually.

### 2.2.3.1.2 Tiles

Following tiles are available and introduced step by step in each world:

- **Ground:** The ground tile allows the player/tank to freely maneuver over it.
- **Wall:** A wall stops all player movement and generally stops all projectiles (see [weapons/enemies](#) for more).
- **Hole:** A hole stops all player/tank movement but does not stop projectiles at all.
- **Switches & Gates:** At first a gate behaves like a wall. When the player drives over a switch the gate opens and starts behaving like a ground tile. This can be reversed by driving over the switch again.
- **Destroyable Walls:** This wall has a certain number of HP. They can be damaged by all projectiles. Once their HP reaches 0 they are destroyed and behave like ground tiles.

### 2.2.3.2 World

Each world contains a certain number of levels. They differentiate between each other by introducing new tiles and enemies.

#### 2.2.3.2.1 Checkpoints

Certain levels can be set to be a checkpoint. Once completing a checkpoint level, the game is saved. When the player dies, he can continue the game from the last checkpoint.

## 2.2.4 UI

### 2.2.4.1 Menu and navigation

The menu allows the player to navigate the game. Following screens are available:

- **Main menu:** This screen has following buttons:
  - **Start/Continue:** When no save-file is detected, the button shows the text **Start**. When clicking it, the player starts a new game and the first level is loaded. When a save-file is detected, the button shows **continue**, allowing the player to continue the game at the last checkpoint saved in the save-file.
  - **Options:** Clicking this button opens the settings menu.
  - **Exit:** Clicking this button exits the game.
- **Settings:** The settings menu can be reached by the main screen and the pause menu. It allows the player to change following settings:
  - **Music volume:** A slider to change the volume of the music
  - **SFX volume:** A slider to change the volume of the SFX
  - **Back:** A button to return to the most recent menu.

- **Shop:** The shop can be used by the player to buy different upgrades and new weapons (see section [Upgrades](#) and [Weapons](#)) after a level is completed.
  - A list is of all upgrades and weapons with their corresponding prices is shown. The play can click the upgrade/wapon to buy and equip it.
  - **Continue:** Clicking this button takes the player to the next level.
- **Pause menu:** The pause menu can be reached in-game by pressing the pause-button. It has following buttons:
  - **Continue:** Continue the game. Pressing the pause button again has the same effect.
  - **Options:** Clicking this button opens the settings menu.
  - **Exit:** Return to main menu. The game is not saved.

#### 2.2.4.2 HUD

While in-game, following information is displayed to the player on the HUD:

- Current HP of the player displayed in a healthbar.
- Number of remaining enemies.
- Number of coins the player is in possession of.
- Current level.

## 2.3. Nonfunctional requirements

### 2.3.1. User interface and human factors

The user interface and gameplay should be self-explanatory and easy to learn. Therefore following points shold be followed:

- Buttons should be placed in plausible positions.
- New gameplay elements are intruduced by a self-explanatory level. Text is only used as a last resort.
- A tutorial is not needed, only the controls are shown in the first level.

### 2.3.2. Documentation

Each step in development should be documented. To achieve this, following points must be considered and implemented:

- Version control
- Task management
- Documents
  - Requirement Analysis
  - Diagrams
  - Report
- Clean, documented and readable code

### 2.3.3. Hardware and performance characteristics

The game is accessible for all windows and linux environments and runs at smooth 60 FPS on low end (to be further defined) hardware.

### 2.3.4. Error handling and extreme conditions

Error handling is important, the game must not crash on any circumstances. If an critical error occurs the player is notified and the system resets to an acceptable state.

#### 2.3.4.1 Error Handling

- Implement boundary checks for tank movement to prevent it from going off the playable area.
- Use exception handling for input validation (e.g., preventing non-numeric inputs).

#### 2.3.4.2 Extreme Conditions

Limit for projectiles fired to prevent overwhelming the game environment by adding a timer for each bullet to get destroyed after the timer runs out. In addition the sps is capped.

### 2.3.5. Quality issues

To ensure code quality, following criteria must be considered:

- **Test Driven Development:** Each function with return parameter or input parameter must be tested by at least one unit test.
- **Documented Code:** Code, which is not easily understandable is commented.
- **Code Review:** In order to perform a pull request into the development branch, at least one other person must review the code and approve the changes.

### 2.3.6. System modifications

Modding support is not officially supported. But there are things you have to stick to

- Use an level designs, allowing seamless addition of new maps without altering existing code extensively.
- Incorporate a flexible tank customization system, enabling the addition of new tank types, weapons or upgrades in future updates.

### 2.3.7 Physical environment

Involves ensuring the game's adaptability across various devices, screen sizes, and performance capabilities:

#### 2.3.7.1 Responsive Design

- **Screen Size Adaptation:** Ensure that game elements, including UI components and gameplay visuals, scale appropriately across different screen sizes and aspect ratios.
- **Aspect Ratio Handling:** Design the game to support multiple aspect ratios without distortion or loss of content, ensuring a consistent visual experience.

### 2.3.7.2 Accessibility Considerations

- **Usability Testing:** Conduct usability tests to ensure that all features are effective and intuitive for users with different abilities.

## 2.4 Pseudo Requirements

Develop pixel art designs that evoke nostalgia and capture the essence of classic tank games. Also ensure that sound effects and background music complement the retro pixel art style, enhancing the overall gaming experience.

## 2.5. System models

see Folder "documentation"

# 3. Glossary

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